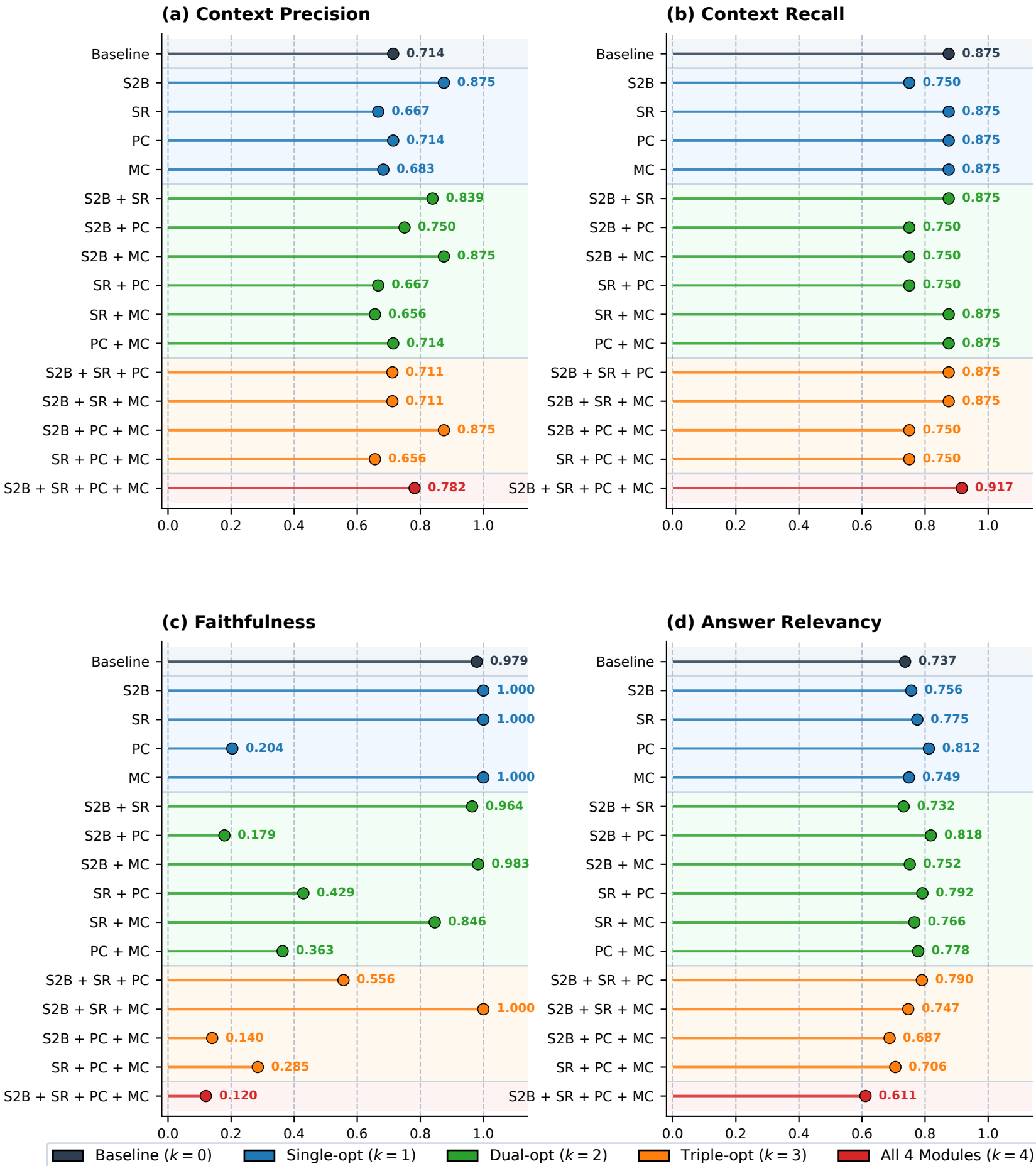
RAGAS Quality Metrics by Pipeline Architecture — Enterprise
(a) Context Precision
Baseline 0.714
S2B 0.875
SR 0.667
PC 0.714
MC 0.683
S2B + SR 0.839
S2B + PC 0.750
S2B + MC 0.875
SR + PC 0.667
SR + MC 0.656
PC + MC 0.714
S2B + SR + PC 0.711
S2B + SR + MC 0.711
S2B + PC + MC 0.875
SR + PC + MC 0.656
S2B + SR + PC + MC 0.782
0.0 0.2 0.4 0.6 0.8 1.0
(b) Context Recall
Baseline 0.875
S2B 0.750
SR 0.875
PC 0.875
MC 0.875
S2B + SR 0.875
S2B + PC 0.750
S2B + MC 0.750
SR + PC 0.750
SR + MC 0.875
PC + MC 0.875
S2B + SR + PC 0.875
S2B + SR + MC 0.875
S2B + PC + MC 0.750
SR + PC + MC 0.750
S2B + SR + PC + MC 0.917
0.0 0.2 0.4 0.6 0.8 1.0
(c) Faithfulness
Baseline 0.979
S2B 1.000
SR 1.000
PC 0.204
MC 1.000
S2B + SR 0.964
S2B + PC 0.179
S2B + MC 0.983
SR + PC 0.429
SR + MC 0.846
PC + MC 0.363
S2B + SR + PC 0.556
S2B + SR + MC 1.000
S2B + PC + MC 0.140
SR + PC + MC 0.285
S2B + SR + PC + MC 0.120
0.0 0.2 0.4 0.6 0.8 1.0
(d) Answer Relevancy
Baseline 0.737
S2B 0.756
SR 0.775
PC 0.812
MC 0.749
S2B + SR 0.732
S2B + PC 0.818
S2B + MC 0.752
SR + PC 0.792
SR + MC 0.766
PC + MC 0.778
S2B + SR + PC 0.790
S2B + SR + MC 0.747
S2B + PC + MC 0.687
SR + PC + MC 0.706
S2B + SR + PC + MC 0.611
0.0 0.2 0.4 0.6 0.8 1.0
Baseline ($k = 0$)
Single-opt ($k = 1$)
Dual-opt ($k = 2$)
Triple-opt ($k = 3$)
All 4 Modules ($k = 4$)